

Atrey Desai

Brendan Iribe Center for Computer Science & Engineering
University of Maryland
College Park, MD 20742

✉ adesai10@umd.edu
🏠 atreydesai.com
🎓 Google Scholar

EDUCATION

University of Maryland, College Park

Exp. Graduation: May 2027

B.S. in Computer Science (Honors) & B.A. in Linguistics (Minor in Korean Studies)

* – in progress

Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber

Coursework: Natural Language Processing (Grad), Commonsense Reasoning (Grad), Mechanistic Interpretability (Grad)*, Machine Learning, Data Science, Parallel Computing*, Computer Systems, Syntax, Semantics*, Phonetics, Psycholinguistics

EXPERIENCE

University of Maryland, College Park

May 2024 – present

Undergrad Researcher, CLIP Lab

Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber

- Quantified differential LLM susceptibility to human-written vs. LLM-generated MCQs to assess if questions contain unintended artifacts and are solvable without full context; created tools to isolate distractor-choice provenance effects and improve the reliability of human synthetic data for model evaluation and distillation.
- Constructed an adversarial benchmark to test VLM capabilities in detecting out-of-context (OOC) video-based misinformation on social media based on multimodal clues and user interactions to probe reasoning under deceptive framing.
- Applied causal interpretability techniques to demonstrate that LMs require significantly more data than humans to acquire general syntactic representations, revealing heightened sensitivity to item-level and construction-level variation.
- Developed an oracle-free framework for estimating process reward annotation quality without ground truth using behavioral signals, inducing better step-by-step reasoning in Process Reward Models (PRMs) for RL in non-verifiable tasks.

The University of Texas at Arlington

Feb 2024 – present

Visiting Researcher, ACL2 Lab & NSF

Advisor: Prof. Kenny Zhu

- Designed AniVoice-cat, a dataset of 26,000+ annotated cat vocalizations from 250+ hours of video and identified 57 unique cat phones, establishing a foundational resource for computational research in non-human communication.
- Improved transcription pipeline using PANNs and HuBERT models to 96% accuracy with 93% top-5 accuracy in action recognition, setting a new state-of-the-art for automated animal vocalization analysis.

Learn Prompting

May 2025 – Jan 2026

Member of Technical Staff (San Francisco, CA)

- Ran Hackaprompt, the world's largest red-teaming hackathon, overseeing user engagement and technical challenges.
- Researched the robustness of AI safety judges against CBRNE-related adversarial content.

University of Maryland, College Park

Dec 2023 – Aug 2024

Researcher, FIRE Sustainability Analytics Lab

Advisor: Prof. Thanicha Ruangmas

- Developed Python pipeline for environmental impact assessments of U.S. emissions, enabling more efficient policy.
- Drafted a framework to guide evidence-based policymaking on climate restoration strategies.

Brown University

Dec 2020 – June 2023

Researcher, Reinforcement Learning at Brown Group

Advisor: Prof. Michael Littman

- Developed a custom RL environment that empowered non-technical users to programmatically solve complex tasks by defining reward functions and specifying agent behavior, reducing task setup time.
- Published and presented research at two top-tier workshops (AAAI, RLDM), demonstrating how human-readable interfaces enable fine-grained control during inference and improve AI-human interaction in robotics.

PUBLICATIONS

Preprints

* – equal contribution

1. A Preview of Computational Animal Linguistics
Under Review (ACM Computing Surveys)

Atrey Desai, Tirza Panunto, Lindsay Pike, Theron S. Wang, Tuan M. Dang, Hridayesh Lekhak and Kenny Q. Zhu

Publications

5. Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?
ACL 2026 (Findings), [Oral at TLS 2026](#)

Atrey Desai and Sathvik Nair

4. BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks
ACL 2026
Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, . . . , Jordan Boyd-Graber
3. Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers
ACL 2026
Nishant Balepur, **Atrey Desai** and Rachel Rudinger
2. Language Models Generate Multiple-Choice Questions with Artifacts
MASC-SLL 2025
Atrey Desai, Nishant Balepur and Rachel Rudinger
1. Reinforcement Learning As End-User Trigger-Action Programming
IML Workshop @ AAAI 2022; RLDM 2022
Chace Hayhurst, Hyoujae Park, **Atrey Desai**, Suheidy De Los Santos and Michael Littman

SELECTED AWARDS AND HONORS

- **SPIRE Research Grant** (\$3,000) 2025
- **Omicron Delta Kappa Top 10 Freshman** 2024
- **CMSC & ARHU Dean's List** 2023 – 2026
- **UMD President's Scholarship** (\$50,000) 2023
- **NMSC National Merit Scholarship** (\$4,000) 2023
- **Catherine Yang Scholarship** (\$1,000) 2023

TALKS

- **Filler-Gap Dependencies under Developmental Constraints in LMs**
 - Texas Linguistics Society Feb 2026
 - UMD Computational Cognitive Science Reading Group Feb 2026
- **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
 - University of Maryland, College Park Nov 2024
 - UT Arlington, Department of Computer Science July 2024

PROFESSIONAL RESPONSIBILITIES

- **Subreviewer**, ACL Rolling Review (ARR) 2025 – present
- **Member**, University Research Advisory Board 2025 – present
- **Vice Chair**, Computer Science Department Council 2025 – present
Appointed by faculty and student body; represent 4,200+ CS undergraduates.
- **Senior Member**, FIRE Student Leadership Council 2024 – present
Represented 1000+ peers, ran events & workshops, and reformed class curricula.
- **University Ambassador** 2024 – present
Represented the CS Dept. & CMNS College at admissions events and to official guests.
- **Mentorship**
 - **Office of Undergraduate Research:** Juan Cortés, Kemisola Benson, Vivian Akpala 2025
 - **Technica Mentoring:** Savya Miriyala, Tanya Grover, Jessica Ononye, Nakshatra Hiray 2024
 - **MSET Robotics:** Workshop organizer and curriculum designer 2020 – 2022

SKILLS

- **Languages:** Python, Java, JavaScript/HTML/CSS, R, MATLAB.
- **Libraries/Frameworks:** Huggingface (Datasets, Transformers), NLTK, PyTorch, Selenium, BeautifulSoup.
- **Tools:** Git, Docker, GCP, Google Vertex AI, VS Code.
- **Natural Languages:** English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner).